

Lecture Notes 5-6, Probability

Continuous distributions

1) *Uniform distribution*: X is uniformly distributed over an interval $[a, b]$ (notation: $X \sim U(a, b)$), if its density function is constant: $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$, and 0 elsewhere. Its (cumulative) distribution function is $F_X(x) = \frac{x-a}{b-a}$ for $x \in [a, b]$, 0 for $x < a$, and 1 for $b < x$.

The common random number generators produce numbers uniformly distributed over $[0, 1]$.

2) *Exponential distribution*: $X \sim \text{Exp}(\lambda)$ if its density function is $f_X(x) = \lambda e^{-\lambda x}$, and its distribution function is $F_X(x) = 1 - e^{-\lambda x}$ (where λ is a positive parameter).

This distribution is the continuous version of Geometric distribution: we are waiting for an event in continuous time. It is said to be "forever young" because the probability of the event (often called "ruin") happening in an interval of fixed length does not change with the progress of time. (Like at the geometric distribution: we always have the same chance to hit the target, at each trial.)

Example: we can suppose that the distribution of miles done (X) till our car gets a tyre puncture is exponential (say, with parameter 10^{-4}). We have to drive 543 miles to Berlin and 543 miles back. What's the probability that we will not get a puncture during the way there? And during the whole journey?

Answer: $P(X \leq 543) = F_X(543)$ is the probability that we are going to get a puncture before reaching Berlin. So not getting a puncture has probability $1 - F_X(543) = 1 - (1 - e^{-\lambda \times 543}) = e^{-0.0543} = 0.947$, about 95%.

$P(X \leq 1086) = F_X(1086)$ is the probability that we will get a puncture somewhere during the whole way. So not getting a puncture has probability $1 - F_X(1086) = 1 - (1 - e^{-\lambda \times 1086}) = e^{-0.1086} = 0.897$, a bit less than 90%.

3) *The Normal (or Gaussian) distribution* is remarkably useful because of the Central Limit Theorem: under mild conditions, it states that *the average of independent random variables* is approximately Normally distributed.

$X \sim N(0, 1)$ (Standard Normal) if its density function is $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$. (For The Standard Normal distribution, the mean is 0 and the variance is 1, these are the parameters given in parenthesis.)

More generally, $X \sim N(\mu, \sigma^2)$ if its density function is $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$. Its distribution function (the usual notation is $\Phi(x)$) does not have a closed form, but tables are available with its values.

A normally distributed random variable can take any values on the real line, but it is concentrated around its mean, and 95% of the values are less than two standard deviations away from the mean. We also know that approximately 68% of the experiments fall within one standard deviation from the mean. This a way for comparing normally distributed random variables: we measure the outcomes in multiples of standard deviation above or below the mean.

As an example, suppose we know that Bill scored 90 in the Economics exam and Linda scored 80 in Law. We know also that, as is often the case, the candidates scores within each subject are approximately Normal in distribution.

Next, we need to ask how the two scores relate to those of other students. Well, the mean score in Economics was 60, and in Law it was 65. Does that make Bill's the better score? - Not quite. We need to know how many other students exceeded the mean by as much as Bill and Linda.

Suppose now I tell you that in Economics the standard deviation was 15 marks, while in Law it was 5 marks. So which student had the better score, compared with his or her fellow candidates? - This means Bill's score of 90 was two standard deviations above the mean, and Linda's score of 80 was three standard deviations above the mean. Clearly, then, Linda has the better score, compared with fellow candidates.

Again, which is the better score, 60 in Economics or 60 in Law? - 60 in Economics is the mean score but, in Law, 60 is one standard deviation below the mean. So 60 in Economics is the better score.

Formulas for the Expectation and Variance of a random variable

Expected value of a discrete random variable:

$$E(X) = \sum_k kP(X = k).$$

Changing the sum to an integral, and the probabilities to density function, the expected value of a continuous random variable is

$$E(X) = \int_{-\infty}^{+\infty} xf_X(x)dx.$$

That is, we calculate the average of the values of X , weighted by their probabilities.

If he have to calculate the expected value of a function of X , similarly, we take the average of the possible values of that function, weighted by the probabilities of those values:

$$E(g(X)) = \sum_k g(k)P(X = k)$$

for discrete variables, and

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x)f_X(x)dx$$

for continuous variables.

For easier calculation of the variance, we need the following important properties of the expected value:

$E(Y_1 + Y_2) = E(Y_1) + E(Y_2)$, and $E(cY) = cE(Y)$ where c is a constant. I could have written

$$E(c_1Y_1 + c_2Y_2) = c_1E(Y_1) + c_2E(Y_2).$$

This property is the *linearity* of expectation. And also

$$E(c) = c$$

if c is a constant.

This leads to a useful transformation, a formula for the variance:

$$Var(X) = E\{[X - E(X)]^2\} = E\{X^2 + [E(X)]^2 - 2XE(X)\} = E(X^2) + [E(X)]^2 - 2E(X)E(X) = E(X^2) - [E(X)]^2.$$

So

$$Var(X) = \sum_k k^2P(X = k) - \left(\sum_k kP(X = k)\right)^2$$

for discrete and

$$Var(X) = \int_{-\infty}^{+\infty} x^2 f_X(x)dx - \left(\int_{-\infty}^{+\infty} xf_X(x)dx\right)^2$$

for continuous random variables.

For example, the expectation of the $U(0, 1)$ distribution is

$$\int_0^1 x dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1^2 - 0^2}{2} = \frac{1}{2}$$

and its variance is

$$\int_0^1 x^2 dx - (E(X))^2 = \left[\frac{x^3}{3} \right]_0^1 - \left(\frac{1}{2} \right)^2 = \frac{1^3 - 0^3}{3} - \frac{1}{4} = \frac{1}{12}.$$

Important property of the variance:

$$Var(aX + b) = Var(aX) = a^2 Var(X),$$

or the equivalent statement using standard deviations:

$$SD(aX + b) = SD(aX) = a SD(X).$$

Now you can check that the expectation of a random variable having general uniform distribution $U(a, b)$ is $E(X) = \frac{a+b}{2}$ and the variance is $Var(X) = \frac{(b-a)^2}{12}$, so its standard deviation is $SD(X) = \frac{b-a}{\sqrt{12}}$.

We don't calculate the expectation and variance of the exponential distribution here but I give them as facts:

If $X \sim Exp(\lambda)$ then $E(X) = \frac{1}{\lambda}$ and $Var(X) = \frac{1}{\lambda^2}$, so its standard deviation is $SD(X) = \frac{1}{\lambda}$.

For the Normal distribution, it is easy to remember its expectation and variance because these are its parameters (in the formula given above for the density function, μ is the expectation and σ^2 is the variance).

An important but advanced theoretical notion is the moment generating function. First we define it for number series in general, then we proceed to random variables.

The generating function of a number series (a_i) , $i = 0, 1, 2, \dots$ is

$$G(s) = \sum_{i=0}^{\infty} a_i s^i.$$

Differentiating it k times, the lower powers of s disappear. Substituting $s = 0$ make the higher power terms disappear, too. So the k 'th derivative at 0, divided by $k!$, gives a_k .

Similarly, we can define the probability generating function of a nonnegative-integer-valued random variable X as

$$G(s) = E(s^X) = \sum_{i=0}^{\infty} s^i P(X = i),$$

and the k 'th derivative, divided by $k!$ and substituted $s = 0$, gives back the probability $P(X = k)$ for each k .

So giving the function $G(s)$ is another way for describing a distribution.

Similarly, we can substitute $s = e^t$, and so $G(s) = E(s^X) = E(e^{tX}) = M(t)$. This can be defined for any real t (by $M(t) = \int e^{tX} dF(x)$, a Stieltjes-integral, this considers any type of distribution functions).

And the last version is the *characteristic function*: $\phi(t) = E(e^{itX})$ with $i = \sqrt{-1}$. $\phi(t)$ is better behaved than $M(t)$.

Giving $\phi(t)$ is equivalent to giving the distribution (there is an inversion formula, called inverse Fourier transform).

One of the advantages of using the characteristic function of random variables instead of their probability or density function is, that the sum of independent random variables can be dealt with much more easily. The density function of $Z = X + Y$ (if X and Y are independent) is the *convolution* of the f_X and f_Y density functions:

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z - x) dx.$$

For characteristic functions, if X and Y are independent, we have just a product instead:

$$\phi_Z(t) = E(e^{it(X+Y)}) = E(e^{itX})E(e^{itY}) = \phi_X(t)\phi_Y(t).$$

This property will be useful in proving the Law of Large Numbers and the Central Limit Theorem.